

Roll number :

**ASSIGNMENT 1**

**Q2) ATM simulation**

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
void checkBalance(double balance)
```

```
{  
    cout << "Current Balance: " << balance << endl;  
}
```

```
void depositMoney(double &balance, string transactions[], int &count)
```

```
{  
    double amount;  
  
    cout << "Enter amount to deposit: ";  
    cin >> amount;  
  
    balance += amount;  
  
    if (count < 100)  
    {  
        transactions[count] = "Deposit";  
        count++;  
    }  
  
    cout << "Deposit successful." << endl;  
}
```

```
void withdrawMoney(double &balance, string transactions[], int &count)
```

```
{  
    double amount;  
  
    cout << "Enter amount to withdraw: ";  
    cin >> amount;  
  
    if (amount <= balance)  
    {  
        balance -= amount;  
    }
```

```

        if (count < 100)
        {
            transactions[count] = "Withdrawal";
            count++;
        }

        cout << "Withdrawal successful." << endl;
    }
    else
    {
        cout << "Insufficient balance." << endl;
    }
}

void viewTransactions(string transactions[], int count)
{
    if (count == 0)
    {
        cout << "No transactions found." << endl;
    }
    else
    {
        cout << "Transaction History" << endl;

        for (int i = 0; i < count; i++)
        {
            cout << i + 1 << ". " << transactions[i] << endl;
        }
    }
}

int main()
{
    string pin = "888888";
    string enteredPin;

    int attempts = 0;
    bool loggedIn = false;

    while (attempts < 3)

```

```

{
    cout << "Enter PIN: ";
    cin >> enteredPin;

    if (enteredPin == pin)
    {
        loggedIn = true;
        break;
    }
    else
    {
        attempts++;
        cout << "Incorrect PIN." << endl;
    }
}

if (!loggedIn)
{
    cout << "Maximum attempts reached." << endl;
    return 0;
}

double balance = 1000;
string transactions[100];
int transactionCount = 0;
int choice;

do
{
    cout << "\n1. Check Balance" << endl;
    cout << "2. Deposit Money" << endl;
    cout << "3. Withdraw Money" << endl;
    cout << "4. View Transactions" << endl;
    cout << "5. Exit" << endl;
    cout << "Enter choice: ";
    cin >> choice;

    switch (choice)
    {
        case 1:
            checkBalance(balance);

```

```

        break;

    case 2:
        depositMoney(balance, transactions, transactionCount);
        break;

    case 3:
        withdrawMoney(balance, transactions, transactionCount);
        break;

    case 4:
        viewTransactions(transactions, transactionCount);
        break;

    case 5:
        cout << "Thank you for using the ATM." << endl;
        break;

    default:
        cout << "Invalid choice." << endl;
    }

} while (choice != 5);

return 0;
}

```

### QUESTION 3

```

#include <iostream>
#include <fstream>
#include <string>

using namespace std;

void showHistory()
{
    int choice;

    do

```

```

{
    cout << "\n===== HISTORY OF C++ =====" << endl;
    cout << "1. Origin of C++" << endl;
    cout << "2. Evolution of C++" << endl;
    cout << "3. C++ Standards" << endl;
    cout << "4. Back" << endl;
    cout << "Enter Choice: ";
    cin >> choice;

    switch(choice)
    {
        case 1:
            cout << "\nC++ was developed by Bjarne Stroustrup at Bell Labs in 1979." <<
endl;
            cout << "It was initially called C with Classes." << endl;
            break;

        case 2:
            cout << "\nC++ evolved from the C programming language." << endl;
            cout << "It added Object-Oriented Programming features." << endl;
            cout << "Its history is influenced by languages such as Simula and C." << endl;
            break;

        case 3:
            cout << "\n1985 - First C++ Release" << endl;
            cout << "1998 - C++98" << endl;
            cout << "2011 - C++11" << endl;
            cout << "2014 - C++14" << endl;
            cout << "2017 - C++17" << endl;
            cout << "2020 - C++20" << endl;
            break;

        case 4:
            break;

        default:
            cout << "Invalid Choice." << endl;
    }

} while(choice != 4);
}

```

```

void displayChapter(string filename)
{
    ifstream file(filename);

    if(!file)
    {
        cout << "Chapter file not found." << endl;
        return;
    }

    string line;

    while(getline(file, line))
    {
        cout << line << endl;
    }

    file.close();
}

void learnCpp()
{
    int choice;

    do
    {
        cout << "\n==== LEARN C++ =====> << endl;
        cout << "1. Data Types" << endl;
        cout << "2. Control Flow" << endl;
        cout << "3. Loops" << endl;
        cout << "4. Arrays" << endl;
        cout << "5. File Streams" << endl;
        cout << "6. Back" << endl;
        cout << "Enter Choice: ";
        cin >> choice;

        switch(choice)
        {
            case 1:
                displayChapter("datatypes.txt");

```

```

        break;

    case 2:
        displayChapter("controlflow.txt");
        break;

    case 3:
        displayChapter("loops.txt");
        break;

    case 4:
        displayChapter("arrays.txt");
        break;

    case 5:
        displayChapter("filestreams.txt");
        break;

    case 6:
        break;

    default:
        cout << "Invalid Choice." << endl;
    }

} while(choice != 6);
}

int main()
{
    int choice;

    do
    {
        cout << "\n===== " << endl;
        cout << "FOUNDATIONS AND HISTORY OF C++" << endl;
        cout << "===== " << endl;
        cout << "1. History of C++" << endl;
        cout << "2. Learn About C++" << endl;
        cout << "3. Exit" << endl;
        cout << "Enter Choice: ";

```

```
cin >> choice;

switch(choice)
{
    case 1:
        showHistory();
        break;

    case 2:
        learnCpp();
        break;

    case 3:
        cout << "Thank you for using the system." << endl;
        break;

    default:
        cout << "Invalid Choice." << endl;
}

} while(choice != 3);

return 0;
}
```